

Non-homogeneous linear higher order differential equations:

$$\frac{d^ny}{dx^n} + p_{n-1}(x) \frac{d^{n-1}y}{dx^{n-1}} + \dots + p_1(x) \frac{dy}{dx} + p_0(x) y = R(x),$$

$R(x) \neq 0$.

$$\Rightarrow \underbrace{\left(D^n + p_{n-1}(x) D^{n-1} + \dots + p_1(x) D + p_0(x) \right)}_{f(D)} y = R(x) \quad \text{--- (1)}$$

Theorem: If $y_1, y_2, \dots, y_n$ are n L.I. solutions of homogeneous linear diff eq $f(D)y = 0$ and y_p is some particular solutions of $f(D)y = R(x)$, then general solution of Eq. (1) is given by

$$y = C_1 y_1 + C_2 y_2 + \dots + C_n y_n + y_p.$$

Proof: Take $z = y - y_p$, y is a solution of $f(D)y = R(x)$.

$$\begin{aligned} \text{Then } f(D)z &= f(D)(y - y_p) = f(D)y - f(D)y_p \\ &= R(x) - R(x) \\ &= 0 \end{aligned}$$

$\Rightarrow z$ is a solution of $f(D)y = 0$.

Therefore, $\exists$ constants $C_1, C_2, \dots, C_n$ s.t

$$z = C_1 y_1 + C_2 y_2 + \dots + C_n y_n$$

$$\Rightarrow y - y_p = C_1 y_1 + C_2 y_2 + \dots + C_n y_n$$

$$\Rightarrow y = \underbrace{C_1 y_1 + C_2 y_2 + \dots + C_n y_n}_{\substack{\text{complementary function} \\ \text{(C.F.)}}} + \underbrace{y_p}_{\substack{\text{particular integral} \\ \text{(P.I.)}}}$$

- Note:**
- ① The general solution of the corresponding homogeneous equation ($f(D)y = 0$) is called complementary function & it is denoted by C.F.
 - ② A particular solution of non-homogeneous eq. ($f(D)y = R(x)$) is called particular integral & it is denoted by y_p or P.I.

Non-homogeneous linear diff-eq. with constant coefficients:

$$\underbrace{(D^n + a_{n-1}D^{n-1} + \dots + a_1D + a_0)}_{\text{"}} y = R(x), \quad a_i \text{'s are constant.}$$

$$f(D)y = R(x).$$

Inverse operator (A method to find PI):

$$\frac{dy}{dx} = f(x) \Rightarrow Dy = f(x)$$

Integrate, $y = \int f(x) dx$ $\Downarrow$
 $y = \frac{1}{D} f(x)$

Define $\boxed{\frac{1}{D} f(x) = \int f(x) dx}$.

$$\left(\frac{dy}{dx} - ay = R(x) \right) \quad (IF = e^{-ax})$$

$$\therefore y = \underbrace{C e^{ax}}_{C.F.} + \underbrace{e^{ax} \int R(x) e^{-ax} dx}_{P.I.}$$

$$\rightarrow (D-a)y = R(x) \Rightarrow (D-a)y_p = R(x)$$

$$\Downarrow$$

$$y_p = \frac{1}{D-a} R(x)$$

Define $\boxed{\frac{1}{D-a} R(x) = e^{ax} \int R(x) e^{-ax} dx.}$

Suppose $f(D)y = R(x)$, , $f(D) = D^n + a_{n-1}D^{n-1} + \dots + a_1D + a_0$
 $= (D-\alpha_1)(D-\alpha_2)\dots(D-\alpha_n)$

$$\Rightarrow (D-\alpha_1)(D-\alpha_2)\dots(D-\alpha_n)y = R(x)$$

$$\therefore y_p = \frac{1}{(D-\alpha_1)(D-\alpha_2)\dots(D-\alpha_n)} R(x)$$

If all $\alpha_1, \alpha_2, \dots, \alpha_n$ are distinct,

$$\frac{1}{(D-\alpha_1)(D-\alpha_2)\dots(D-\alpha_n)} = \frac{A_1}{D-\alpha_1} + \dots + \frac{A_n}{D-\alpha_n}.$$

Some special cases:

$$(1) f(D)y = e^{ax}$$

$$P.I = \frac{1}{f(D)} e^{ax} = \frac{1}{f(a)} e^{ax}, \quad f(a) \neq 0.$$

$$= x \frac{1}{f'(a)} e^{ax}, \quad f(a) = 0, f'(a) \neq 0.$$

$$= x^2 \frac{1}{f''(a)} e^{ax}, \quad f(a) = 0, f'(a) = 0, f''(a) \neq 0$$

⋮

$$(2) f(D)y = \sin(ax+b) \text{ or } \cos(ax+b)$$

$$P.I = \frac{1}{f(D)} \sin(ax+b) = \frac{1}{f(D^2 = -a^2)} \sin(ax+b),$$

$$f(D^2 = -a^2) \neq 0$$

$$= x \frac{1}{f'(D^2 = -a^2)} \sin(ax+b),$$

$$| \quad f(D^2 = -a^2) = 0,$$

$$| \quad f'(D^2 = -a^2) \neq 0.$$

⋮
similarly for $\cos(ax+b)$.

$$(3) f(D)y = e^{ax} \cdot V, \quad V \rightarrow \text{function of } x$$

$$P.I = \frac{1}{f(D)} e^{ax} \cdot V = e^{ax} \frac{1}{f(D+a)} V.$$

$$(4) f(D)y = x^n$$

$$P.I = \frac{1}{f(D)} x^n, \quad \text{where } f(D) = a_0 D^\beta (1 \pm \phi(D))$$

$$\beta \geq 0, \beta \text{ integer.}$$

$$= \frac{1}{a_0 D^\beta (1 \pm \phi(D))} x^n$$

$$= \frac{1}{a_0 D^\beta} (1 \pm \phi(D))^{-1} x^n.$$

↓

(Use the binomial exp $(1-r)^{-1} = 1+r+r^2+r^3+\dots$).

Ex. Find the general solution of $(D^2-3)y = e^{3x}$.

Sol: Put $y = e^{mx}$ in $(D^2-3)y = 0$. This gives $m^2-3=0 \Rightarrow m=\pm\sqrt{3}$.

$$C.F. = c_1 e^{\sqrt{3}x} + c_2 e^{-\sqrt{3}x}.$$

$$P.I. = \frac{1}{D^2-3} e^{3x} = \frac{1}{3^2-3} e^{3x} = \frac{1}{6} e^{3x}$$

$$\therefore \text{General solution } y = c_1 e^{\sqrt{3}x} + c_2 e^{-\sqrt{3}x} + \frac{1}{6} e^{3x}.$$

Ex. Find the general solution of $y'' - 2y' - 3y = 8e^x$.

Sol: We can write it as $(D^2-2D-3)y = 8e^x$.

C.F.: Put $y = e^{mx}$ in $(D^2-2D-3)y = 0$.

$$\text{This gives, } m^2 - 2m - 3 = 0.$$

$$\Rightarrow (m+1)(m-3) = 0 \Rightarrow m = -1, 3.$$

$$\therefore C.F. = c_1 e^{-x} + c_2 e^{3x}.$$

$$P.I. = \frac{1}{D^2-2D-3} 8e^x = 8 \frac{1}{1^2-2 \times 1-3} e^x = \frac{8}{-4} e^x = -2e^x.$$

$$\therefore \text{General solution } y = C.F. + P.I. \\ = c_1 e^{-x} + c_2 e^{3x} - 2e^x.$$

Ex. Find the general solution of $(D^3-2D^2+D)y = 5e^x$.

Sol: Put $y = e^{mx}$ in $(D^3-2D^2+D)y = 0$.

$$\text{This gives, } m^3 - 2m^2 + m = 0$$

$$\Rightarrow m(m-1)^2 = 0 \Rightarrow m = 0, 1, 1.$$

$$C.F. = c_1 + c_2 e^x + c_3 x e^x.$$

$$P.I. = \frac{1}{D^3-2D^2+D} 5e^x \quad (1^3-2 \times 1 + 1 = 0) \\ = 5x \frac{1}{3D^2-4D+1} e^x \quad (3 \times 1^2 - 4 \times 1 + 1 = 0) \\ = 5x^2 \frac{1}{6D-4} e^x = 5x^2 \frac{1}{6 \times 1 - 4} e^x = \frac{5}{2} x^2 e^x.$$

$$\therefore \text{General sol } y = c_1 + c_2 e^x + c_3 x e^x.$$

Ex. Find the general solution of $(D^2+D+1)y = \sin 2x$.

sol: Put $y = e^{mx}$ in $(D^2 + D + 1)y = 0$.

$$\Rightarrow m^2 + m + 1 = 0$$

$$\Rightarrow m = \frac{-1 \pm i\sqrt{3}}{2}$$

$$\therefore \text{C.F.} = c_1 e^{-\frac{1}{2}x} \cos \frac{\sqrt{3}}{2}x + c_2 e^{-\frac{1}{2}x} \sin \frac{\sqrt{3}}{2}x.$$

$$\begin{aligned} \text{P.I.} &= \frac{1}{D^2 + D + 1} \sin 2x = \frac{1}{-2^2 + D + 1} \sin 2x \\ &= \frac{1}{D - 3} \sin 2x \quad (D^2 \rightarrow -2^2) \\ &= \frac{D + 3}{D^2 - 9} \sin 2x = \frac{D + 3}{-2^2 - 9} \sin 2x \\ &= -\frac{1}{13} (D \sin 2x + 3 \sin 2x) \\ &= -\frac{1}{13} (2 \cos 2x + 3 \sin 2x) \end{aligned}$$

$$\therefore \text{General sol } y = \text{C.F.} + \text{P.I.}$$

Ex. find the general solution of $(D^3 + 9D)y = \cos 3x$.

sol: Put $y = e^{mx}$ in $(D^3 + 9D)y = 0$

$$\Rightarrow m^3 + 9m = 0$$

$$\Rightarrow m = 0, \pm 3i.$$

$$\therefore \text{C.F.} = c_1 + c_2 \cos 3x + c_3 \sin 3x.$$

$$\begin{aligned} \text{PI} &= \frac{1}{D^3 + 9D} \cos 3x \\ &= x \frac{1}{3D^2 + 9} \cos 3x \quad (D^2 \rightarrow -3^2) \\ &= x \frac{1}{3(-3^2) + 9} \cos 3x \quad (-9D + 9D = 0) \\ &= -\frac{x}{18} \cos 3x. \end{aligned}$$

$$\text{General sol } y = \text{C.F.} + \text{P.I.}$$

Ex. Find the general solution of $(D^2 + 3D - 4)y = x^2$.

sol: Put $y = e^{mx}$ in $(D^2 + 3D - 4)y = 0$.

$$\Rightarrow m^2 + 3m - 4 = 0$$

$$\Rightarrow (m-1)(m+4) = 0$$

$$\Rightarrow m = 1, -4.$$

$$C.F. = C_1 e^x + C_2 e^{-4x}.$$

$$P.I. = \frac{1}{D^2+3D-4} x^2$$

$$= -\frac{1}{4} \frac{1}{\left(1 - \frac{D(D+3)}{4}\right)} x^2$$

$$= -\frac{1}{4} \left(1 - \frac{D(D+3)}{4}\right)^{-1} x^2$$

$$= -\frac{1}{4} \left(1 + \frac{D(D+3)}{4} + \frac{D^2(D+3)^2}{4^2} + \dots\right) x^2$$

$$= -\frac{1}{4} \left(x^2 + \frac{D(D+3)}{4} x^2 + \frac{D^2(D+3)^2}{16} x^2 + 0\right)$$

$$= -\frac{1}{4} \left(x^2 + \frac{1}{4}(2+3 \times 2x) + \frac{9}{16} x^2\right) \quad (\because D^3 x^2 = 0)$$

$$= -\frac{1}{4} \left(x^2 + \frac{3}{2}x + \frac{1}{2} + \frac{9}{8}\right)$$

General sol $y = C.F. + P.I.$

Ex. Find the general solution of
 $(D^2+4D+4)y = e^{-x} \sin x$

Sol: Put $y = e^{mx}$ in $(D^2+4D+4)y = 0$.

$$\Rightarrow m^2+4m+4 = 0 \Rightarrow (m+2)^2 = 0$$

$$\Rightarrow m = -2, -2.$$

$$C.F. = C_1 e^{-2x} + C_2 x e^{-2x}$$

$$P.I. = \frac{1}{D^2+4D+4} e^{-x} \sin x = \frac{1}{(D+2)^2} e^{-x} \sin x$$

$$= e^{-x} \frac{1}{(D-1+2)^2} \sin x = e^{-x} \frac{1}{(D+1)^2} \sin x$$

$$= e^{-x} \frac{1}{D^2+2D+1} \sin x$$

$$= e^{-x} \frac{1}{-1^2+2D+1} \sin x = e^{-x} \frac{1}{2D} \sin x$$

$$= -\frac{e^{-x} \cos x}{2}.$$

$$\begin{aligned} \text{General solution } y &= \text{C.F.} + \text{P.I.} \\ &= c_1 e^{2x} + c_2 x e^{2x} - \frac{1}{2} e^x \cos x. \end{aligned}$$

Ex. Find the general solution of $(D^4 - 4)y = e^{\sqrt{2}x} \cdot x$

Ex. Find the general solution of $(D^2 + 3D + 2)y = x \sin 2x$.
Hint: For P.I., consider $\sin 2x = \operatorname{Re}(e^{i2x})$